

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2023-2024
FULL MARKS: 75

Math 4641: Numerical Methods

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) What are the steps of problem-solving in engineering? With a suitable example, briefly explain how numerical methods are used to solve an engineering problem.

4
(CO2)
(PO1)

Solution:

The steps of problem-solving in engineering are —

1. Describing or formulating the problem.
2. Using a mathematical model to represent the problem.
3. Solving the mathematical model.
4. Using the solution of the mathematical model to solve the original problem.

e.g. Let's consider a kinematics problem. Given the initial velocity u and the height h of a falling body, we can calculate the time t it needs to reach the ground by finding the positive root of the equation $ut + \frac{1}{2}gt^2 - h = 0$ using a root-finding algorithm.

- b) Derive the formula for the False Position method geometrically.

6
(CO4)
(PO1)

Solution:

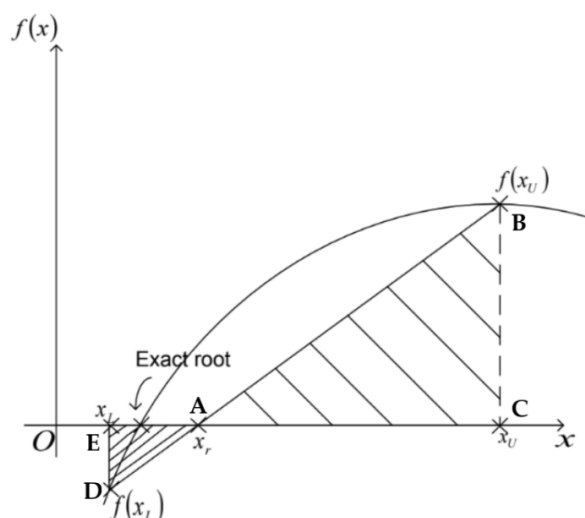


Figure 1: False Position formula derivation.

$$\triangle ABC \sim \triangle DAE$$

So,

$$\frac{DE}{AE} = \frac{BC}{AC}$$

$$\frac{0 - f(x_L)}{x_r - x_L} = \frac{0 - f(x_U)}{x_r - x_U}$$

From the above equation, one obtains

$$(x_r - x_L)f(x_U) = (x_r - x_U)f(x_L)$$

$$x_U f(x_L) - x_L f(x_U) = x_r \{f(x_L) - f(x_U)\}$$

The above equation can be solved to obtain the next predicted root x_r as

$$x_r = \frac{x_U f(x_L) - x_L f(x_U)}{f(x_L) - f(x_U)}$$

The above equation, through simple algebraic manipulations,

$$x_r = x_U - \frac{f(x_U)}{\left\{ \frac{f(x_L) - f(x_U)}{x_L - x_U} \right\}}$$

$$x_r = x_L - \frac{f(x_L)}{\left\{ \frac{f(x_U) - f(x_L)}{x_U - x_L} \right\}}$$

Rubric:

- 4 points for a correct geometric derivation.
- 2 points for the correct formula.

- c) A *monotone function* is a function whose derivative never changes sign, i.e., the function is either always increasing or always decreasing. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a smooth monotone function with a root $x^* = k$, will the Newton–Raphson method always converge to this root, for any choice of the initial guess x_0 ? Justify your answer with a valid proof or a counter-example.

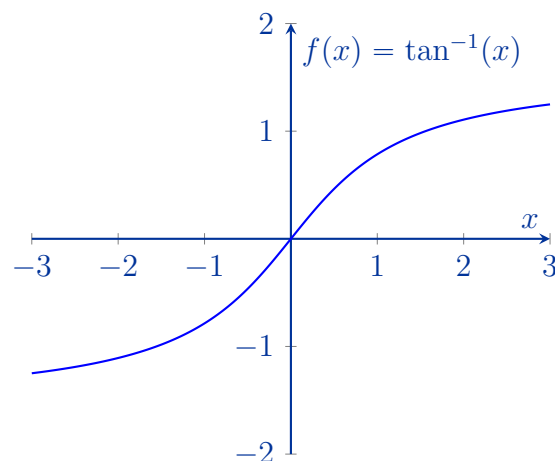
5
(CO3)
(PO1)

Solution:

The Newton–Raphson method will **not** always converge to the root x^* for any initial guess $x_0 \in \mathbb{R}$. A good counter-example to support this claim can be the function

$$f(x) = \tan^{-1}(x)$$

which can be graphically depicted as follows,



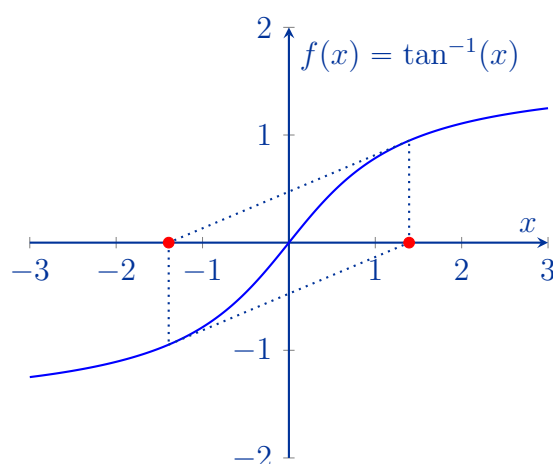
Now, if we start off with an initial guess x_0 that is close to the inflection point (which is also the root in this case), then we will see that the Newton–Raphson method will start to diverge away from the root after each iteration.

Rubric:

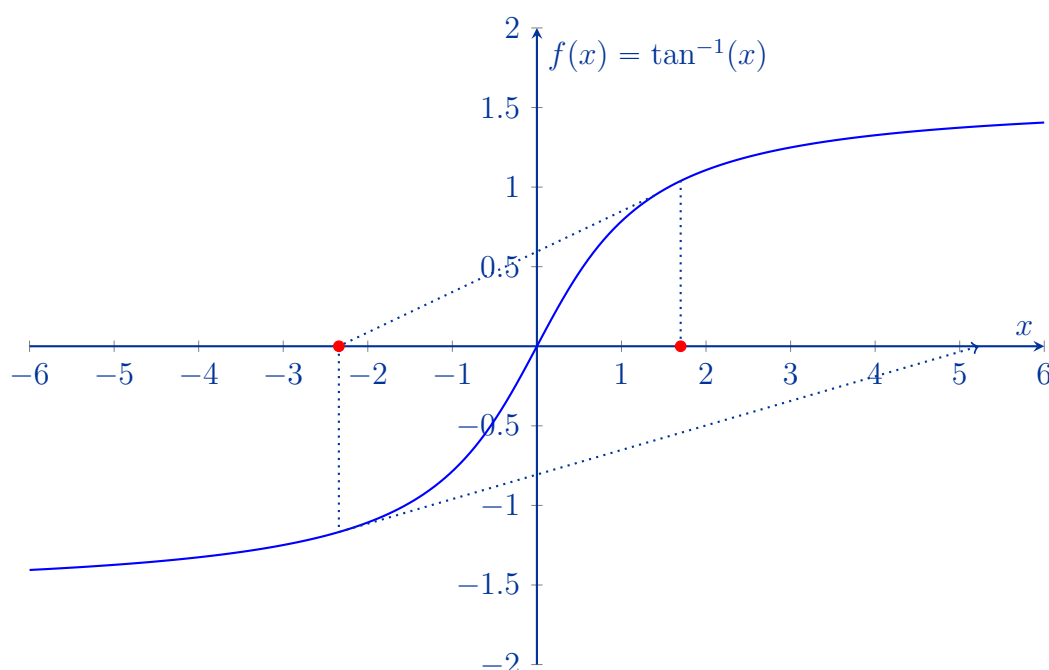
- 1 point for the correct verdict.
- 4 points for a valid counter-example function $f(x)$ and a valid initial guess x_0 .

Elaborate explanation:

It is obvious that a monotone function has a single root. Now, if that function has an inflection point or a platonic portion (also known as a saddle point) where $f'(x) = 0$, then the Newton–Raphson method runs into trouble. More specifically for the aforementioned counter-example function $f(x) = \tan^{-1}(x)$, it can be mathematically shown that if the starting guess x_0 is such that $|x_0| = 1.3917 \dots$, then the algorithm gets stuck in an infinite loop and never converges to the root $x^* = 0$.



And, even worse, if the initial guess x_0 is such that $|x_0| > 1.3917 \dots$, then the Newton–Raphson method gradually diverges from the root without any recourse.



- d) The *Bronze Ratio*, often denoted by the Greek letter β (beta), is the ratio of two quantities a and b , such that $a > b > 0$, if it satisfies the equality $\frac{3a+b}{a} = \frac{a}{b}$. The constant β is equal to the positive root of a particular function $f(x)$. The numerical value of that root is,

$$x_{root}^+ = \beta = \frac{3 + \sqrt{13}}{2} = 3.302775638 \dots$$

- i. From the given scenario, formulate the function $f(x)$.

3
(CO1)
(PO2)

Solution:

According to the given condition,

$$\begin{aligned} \frac{3a+b}{a} &= \frac{a}{b} = \beta \\ \Rightarrow \frac{3a}{a} + \frac{b}{a} - \frac{a}{b} &= 0 \quad \Rightarrow \quad 3 + \frac{1}{\beta} - \beta = 0 \quad \Rightarrow \quad 3\beta + 1 - \beta^2 = 0 \\ \Rightarrow \beta^2 - 3\beta - 1 &= 0 \\ \Rightarrow x^2 - 3x - 1 &= 0 \end{aligned}$$

So, the function is $f(x) = x^2 - 3x - 1$.

- ii. Apply the False Position method, to estimate the value of the bronze ratio β . The initial guesses are $x_l = 1.5$ and $x_u = 4$. Demonstrating the step-by-step mathematical procedure, conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m) at the end of each iteration. Draw Table 1 on your answer script and fill it out after performing the necessary calculations.

11
(CO2)
(PO1)

Table 1: The relevant values obtained in the answer of Question 1.d)ii

Iteration	x_l	x_u	x_r	$ \epsilon_a \%$	m	$f(x_r)$
1	1.5	4		—	—	
2						
3						
4						

Solution:

After performing the series of calculations, we end up with the following table.

Iteration	x_l	x_u	x_r	$ \epsilon_a \%$	m	$f(x_r)$
1	1.5	4	2.8	—	—	−1.56
2	2.8	4	3.2105	12.79	0	−0.3242
3	3.2105	4	3.2875	2.34	1	−0.0548
4	3.2875	4	3.3003	0.39	2	−0.0089

Rubric:

- 0.5 point for each correct answer.
- 1 grace point for demonstrating the step-by-step procedure in each iteration.

2. a) “The magnitude of the true error and the magnitude of the approximate error do not necessarily reflect how bad the error is.” — Do you agree? Justify your answer.

4
(CO3)
(PO1)

Solution:

Yes, I agree with the given statement.

Let's consider an arbitrary function $f_1(x)$. Suppose we get a true error value of $E_t = p$, where p is a small number, while computing the value of $f_1'(2)$ with $h = 0.3$. Now, if we try to do the same thing for another function $f_2(x)$, where $f_2(x) = 10^{-6} \times f_1(x)$, we will get an even more minuscule true error value of $E_t = p \times 10^{-6}$. Even though the problems are the same in essence, we are getting a smaller true error value in the 2nd case, which is giving us an improper interpretability or a false impression of the goodness of the approximation. The same argument holds true for the approximate error as well.

- b) Suppose that, $x = 1.234 \times 10^{-3}$ and $y = 1.225 \times 10^{-3}$. Determine the *minimum* number of significant digits in each of these numerical values:

2
(CO2)
(PO1)

i. $x + y$

ii. $x - y$

iii. xy

iv. $\frac{x}{y}$

Solution:

i $x + y = 2.459 \times 10^{-3}$ has 4 significant digits.

ii $x - y = 0.009 \times 10^{-3}$ has 1 significant digit.

iii $xy = 1.511650 \times 10^{-6}$ has 7 significant digits.

iv $\frac{x}{y} = 1.007346939$ has 10 significant digits.

Rubric:

- 0.5 point for each correct answer.

- c) What is a transcendental function?

1
(CO2)
(PO1)

Solution:

Transcendental functions are functions that cannot be expressed using a finite sequence of terms or algebraic operations.

Rubric:

- 1 point for a correct definition.

- d) Euler's identity, which is $e^{i\pi} + 1 = 0$, is sometimes referred to as the most beautiful equation in mathematics. It is a special case of Euler's formula

$$e^{ix} = \cos(x) + i \sin(x) \quad (1)$$

when evaluated for $x = \pi$. One of the many proofs of this formula involves utilizing the Taylor expansions of its constituent transcendental functions.

The Taylor series for a function $f(x)$ is:

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + f^{(5)}(x)\frac{h^5}{5!} + \dots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of:

- i. e^x ii. $\cos(x)$ iii. $\sin(x)$

Then use these three series to prove Euler's formula as delineated in Equation 1.

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.

We know, $\frac{d}{dx}(e^x) = e^x$, $\frac{d}{dx}(\sin(x)) = \cos(x)$, and $\frac{d}{dx}(\cos(x)) = -\sin(x)$.

- (i) Now, for $f(x) = e^x$, let's calculate the derivatives at the point $x = 0$.

$$f(x) = e^x, f(0) = e^0 = 1; f'(x) = e^x, f'(0) = e^0 = 1; f''(x) = e^x, f''(0) = e^0 = 1; f'''(x) = e^x, f'''(0) = e^0 = 1; f^{(4)}(x) = e^x, f^{(4)}(0) = e^0 = 1;$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$f(0+h) = f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f^{(4)}(0)\frac{h^4}{4!} + \dots$$

$$\Rightarrow f(h) = 1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \frac{h^4}{4!} + \dots$$

$$\Rightarrow f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\Rightarrow e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

- (ii) Now, for $f(x) = \cos(x)$, let's calculate the derivatives at the point $x = 0$.

$$f(x) = \cos(x), f(0) = \cos(0) = 1; f'(x) = -\sin(x), f'(0) = -\sin(0) = 0; f''(x) = -\cos(x), f''(0) = -\cos(0) = -1; f'''(x) = \sin(x), f'''(0) = \sin(0) = 0; f^{(4)}(x) = \cos(x), f^{(4)}(0) = \cos(0) = 1;$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$\begin{aligned}
 f(0+h) &= f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f^{(4)}(0)\frac{h^4}{4!} + \dots \\
 \Rightarrow f(h) &= 1 + 0 - \frac{h^2}{2!} + 0 + \frac{h^4}{4!} - \dots \\
 \Rightarrow f(h) &= 1 - \frac{h^2}{2!} + \frac{h^4}{4!} - \dots \\
 \Rightarrow f(x) &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \\
 \Rightarrow \cos(x) &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots
 \end{aligned}$$

(iii) Now, for $f(x) = \sin(x)$, let's calculate the derivatives at the point $x = 0$.

$f(x) = \sin(x)$, $f(0) = \sin(0) = 0$; $f'(x) = \cos(x)$, $f'(0) = \cos(0) = 1$; $f''(x) = -\sin(x)$, $f''(0) = -\sin(0) = 0$; $f'''(x) = -\cos(x)$, $f'''(0) = -\cos(0) = -1$; $f^{(4)}(x) = \sin(x)$, $f^{(4)}(0) = \sin(0) = 0$; Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$\begin{aligned}
 f(0+h) &= f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f^{(4)}(0)\frac{h^4}{4!} + \dots \\
 \Rightarrow f(h) &= 0 + h + 0 - \frac{h^3}{3!} + 0 + \dots \\
 \Rightarrow f(h) &= h - \frac{h^3}{3!} + \dots \\
 \Rightarrow f(x) &= x - \frac{x^3}{3!} + \dots \\
 \Rightarrow \sin(x) &= x - \frac{x^3}{3!} + \dots
 \end{aligned}$$

Now,

$$\text{L.H.S} = e^{ix}$$

$$\begin{aligned}
 &= 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \frac{x^4}{4!} + \dots \quad [\text{Since, } i^2 = -1, i^3 = -i, \text{ and } i^4 = 1.] \\
 &= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right) + i\left(x - \frac{x^3}{3!} + \dots\right) \quad [\text{Separating the real and complex terms.}] \\
 &= \cos(x) + i\sin(x) \\
 &= \text{R.H.S.} \quad \square \text{ Q.E.D.}
 \end{aligned}$$

Rubric:

- 4 points for the correct series of e^x .
- 4 points for the correct series of $\cos(x)$.
- 4 points for the correct series of $\sin(x)$.
- 3 points for the correct proof of Euler's formula.

3. a) Why do we use polynomials as interpolants? Briefly describe the properties of polynomials for which they are used as interpolants in numerical analysis.

3
(CO3)
(PO1)

Solution:

Polynomials are used as interpolants because they are easy to —

1. Differentiate
2. Integrate
3. Evaluate

The general template of an n th order polynomial is,

$$P(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_nx^n$$

where a_i are the constant coefficients. We can differentiate $P(x)$ by simply applying the formula $\frac{d}{dx}(a_ix^i) = i \times a_ix^{i-1}$ on each term. We can integrate $P(x)$ by simply applying the formula $\int a_ix^i dx = a_i \frac{x^{i+1}}{i+1}$ on each term. The evaluation process of $P(x)$ is also trivial since it involves plugging in a particular value of x .

Rubric:

- 1 point for discussion about each property.

- b) Suppose, there is an unknown function $f(x)$ and you are given 3 points $(x_0, f(x_0))$, $(x_1, f(x_1))$, and $(x_2, f(x_2))$ through which $f(x)$ passes. Now, to approximate $f(x)$, you decide to use Newton's Divided Difference method of interpolation and obtain a quadratic interpolant $f_2(x)$. Derive the representation of 2nd order Newton's Divided Difference polynomial $f_2(x)$.

8
(CO4)
(PO1)

Solution:



Just [click here](#).

Rubric:

- 1 point for writing the quadratic NDD interpolant $f_2(x)$ in the proper form.
- 1 point for correct derivation of b_0 .
- 2 point for correct derivation of b_1 .
- 3 points for correct derivation of b_2 .
- 1 point for writing the correct expanded form of the quadratic NDD interpolant $f_2(x)$.

- c) The specific heat capacity C of a substance is defined as the amount of heat that is required to raise the temperature of unit mass of that particular substance by 1 degree. Suppose, you are conducting an experiment to determine how much heat is required to bring some volume of water to its boiling point. The values of specific heat capacity C of water that you have calculated at different temperature values T are shown in Table 2.

Table 2: Specific heat C of water as a function of temperature T for Question 3.c

Temperature, T ($^{\circ}\text{C}$)	22	42	52	82	100
Specific Heat, C ($\text{Jkg}^{-1}^{\circ}\text{C}^{-1}$)	4181	4179	4186	4199	4217

Determine the value of the specific heat at $T = 61^{\circ}\text{C}$ using the Direct method of interpolation and a second order polynomial.

Solution:

For second order polynomial interpolation (also called quadratic interpolation), the equation for the specific heat would be —

$$C(T) = a_0 + a_1T + a_2T^2$$

Here, the order $n = 2$. Now, let's choose the nearest $n + 1 = 2 + 1 = 3$ points that bracket the point $T = 61^{\circ}\text{C}$ and evaluate the polynomial. The 3 points are $T_0 = 42$, $T_1 = 52$, and $T_2 = 82$. The respective specific heat values are $C(T_0) = 4179$, $C(T_1) = 4186$, and $C(T_2) = 4199$. Using these values, we obtain the following equations —

$$C(42) = a_0 + a_1 \times 42 + a_2 \times (42)^2 = 4179$$

$$C(52) = a_0 + a_1 \times 52 + a_2 \times (52)^2 = 4186$$

$$C(82) = a_0 + a_1 \times 82 + a_2 \times (82)^2 = 4199$$

Writing these equations in matrix form, we have —

$$\begin{aligned} \begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix} \cdot \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} \frac{533}{50} & -\frac{287}{25} & \frac{91}{50} \\ -\frac{67}{200} & \frac{31}{75} & -\frac{47}{600} \\ \frac{1}{400} & -\frac{1}{300} & \frac{1}{1200} \end{bmatrix} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} 4135 \\ 1.3267 \\ -6.6667 \times 10^{-3} \end{bmatrix} \end{aligned}$$

Hence,

$$C(T) = 4135 + 1.3267T - 6.6667 \times 10^{-3}T^2; \quad 42 \leq T \leq 82$$

At the point $T = 61$, we get,

$$C(61) = 4135 + 1.3267 \times 61 - 6.6667 \times 10^{-3} \times (61)^2 = 4191.2 \text{ Jkg}^{-1}^{\circ}\text{C}^{-1}$$

Rubric:

- 1 point for the correct polynomial representation.
- 1 point for choosing the correct nearest points.
- 2 points for the correct coefficient values.
- 2 points for the correct answer.

d) i. What is the Runge phenomenon?

2
(CO3)
(PO1)

Solution:

Let's consider Runge's function $f(x) = \frac{1}{1 + 25x^2}$ to explain the phenomenon. When we try to approximate the nature of this function using a high-order interpolant, we observe an unhinged oscillatory behavior at either end of the interpolant.

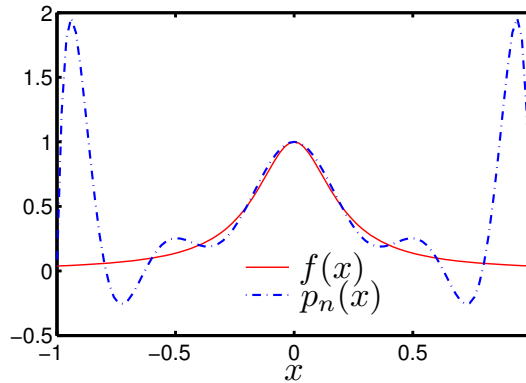


Figure 2: A 9th order polynomial interpolant passing through 10 equidistant points on Runge's function (**badly**).

This is known as Runge phenomenon.

Rubric:

- 1 point for the correct definition.
- 1 point for an example.

ii. A natural cubic spline $S_3(x)$ on $[0, 2]$ is defined by:

5
(CO2)
(PO1)

$$S_3(x) = \begin{cases} 1 + 2x - x^3; & \text{if } 0 \leq x \leq 1, \\ 2 + a(x-1) + b(x-1)^2 + c(x-1)^3; & \text{if } 1 \leq x \leq 2. \end{cases}$$

Given that $S_3(x)$ is a natural cubic spline, it must satisfy the following conditions:

- $S_3(x)$ is continuous at the interior point $x = 1$,
- The second derivative $S_3''(x)$ vanishes at the exterior points $x = 0$ and $x = 2$, as it exhibits no curvature at those points.

Find the values of the constants a , b , and c .

Solution:

Since $S_3(x)$ is a natural cubic spline, it must satisfy the following:

- Continuity of $S_3(x)$, $S_3'(x)$, and $S_3''(x)$ at $x = 1$
- $S_3''(0) = 0$ and $S_3''(2) = 0$ (natural spline boundary conditions)

Matching the first derivatives at $x = 1$:

$$\begin{aligned} \left. \frac{d}{dx}(1 - 2x - x^3) \right|_{x=1} &= \left. \frac{d}{dx}(2 + a(x-1) + b(x-1)^2 + c(x-1)^3) \right|_{x=1} \\ -2 - 3x^2 \Big|_{x=1} &= a + 2b(x-1) + 3c(x-1)^2 \Big|_{x=1} \\ -2 - 3(1)^2 &= a + 0 + 0 \Rightarrow -5 = a \Rightarrow \boxed{a = -5} \end{aligned}$$

Matching second derivatives at $x = 1$:

$$\begin{aligned}\frac{d^2}{dx^2}(1 - 2x - x^3)\Big|_{x=1} &= \frac{d^2}{dx^2}(2 + a(x-1) + b(x-1)^2 + c(x-1)^3)\Big|_{x=1} \\ -6x\Big|_{x=1} &= 2b + 6c(x-1)\Big|_{x=1} \\ -6 &= 2b + 0 \Rightarrow \boxed{b = -3}\end{aligned}$$

Natural spline condition: $S''(2) = 0$

$$S''(x) = 2b + 6c(x-1) \Rightarrow S''(2) = 2b + 6c(1) = 0$$

$$2(-3) + 6c = 0 \Rightarrow -6 + 6c = 0 \Rightarrow \boxed{c = 1}$$

Therefore, the final answers are,

$$\boxed{a = -5, \quad b = -3, \quad c = 1}$$

Rubric:

- 2 points for the correct value of a .
- 2 points for the correct value of b .
- 1 points for the correct value of c .